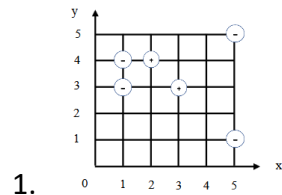


CS 5990 - Assignment #4

Bronco ID: 015765555

Last Name: Prasetyo

First Name: Fidelis



a. 1NN

x	y	Class	1 st NN class	Correct?
1	3	-	-	Yes
1	4	-	- (tie)	No
2	4	+	-	No
3	3	+	+	Yes
5	1	-	+	No
5	5	-	+	No

$$\text{error rate} = \frac{4}{6} = 0.667$$

b. 3NN

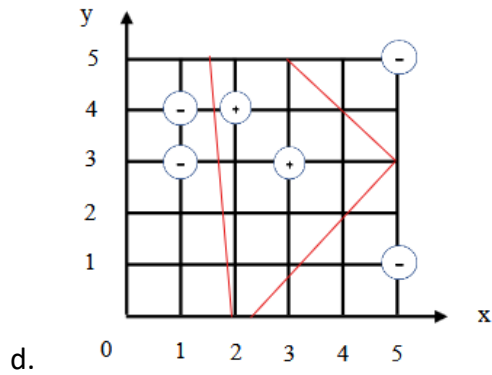
x	y	Class	1 st NN class	2 nd NN class	3 rd NN class	Correct?
1	3	-	-	+	+	No
1	4	-	-	+	+	No
2	4	+	-	-	+	No
3	3	+	+	-	-	No
5	1	-	+	-	+	No
5	5	-	+	+	-	No

$$\text{error rate} = \frac{6}{6} = 1$$

c. 3NN(DW)

x	y	Class	1 st NN class	2 nd NN class	3 rd NN class	3NN(DW) class	Correct?
1	3	-	- (1)	+(0.5)	+(0.25)	-	Yes
1	4	-	- (1)	+(1)	+(0.2)	+	No
2	4	+	- (1)	-(0.5)	+(0.5)	-	No
3	3	+	+(0.5)	-(0.25)	-(0.2)	+	Yes
5	1	-	+(0.125)	-(0.0625)	+(0.0556)	+	No
5	5	-	+(0.125)	+(0.1)	-(0.0625)	+	No

$$\text{error rate} = \frac{4}{6} = 0.667$$



2. KNN

	t1	t2	t3	t4	t5	Class
P1	2	3	4	5	5	Buy
P2	0	1	2	3	5	Buy
P3	2	2	2	2	4	Sell
P4	3	3	3	3	3	Sell
P5	4	2	4	4	4	Sell
s	1	2	3	4	4	?

$$d(s, P1) = 1 + 1 + 1 + 1 + 1 = 5$$

$$d(s, P2) = 1 + 1 + 1 + 1 + 1 = 5$$

$$d(s, P3) = 1 + 0 + 1 + 2 + 0 = 4$$

$$d(s, P4) = 2 + 1 + 0 + 1 + 1 = 5$$

$$d(s, P5) = 3 + 0 + 1 + 0 + 0 = 4$$

$$\text{Class}(s) = \text{Sell}$$

3. <https://github.com/fidelisprasetyo/CS5990/tree/hw4>

Record	A	B	C	Class
1	0	0	0	+
2	0	0	1	-
3	0	1	1	-
4	0	1	1	-
5	0	0	1	+
6	1	0	1	+
7	1	0	1	-
8	1	0	1	-
9	1	1	1	+
10	1	0	1	+

4.

$$P(\text{Class} = + | A = 0, B = 1, C = 0) =$$

$$(P(A = 0 | \text{Class} = +) * P(B = 1 | \text{Class} = +) * P(C = 0 | \text{Class} = +)) * P(\text{Class} = +) =$$

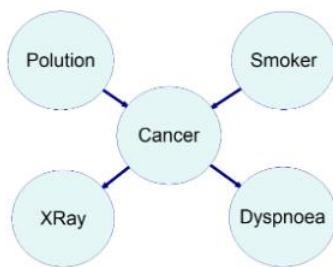
$$\frac{2}{5+4} * \frac{1+2}{5+4} * \frac{1+2}{5+4} * \frac{5}{10} = 0.025$$

$$P(\text{Class} = - | A = 0, B = 1, C = 0) = \\ (P(A = 0 | \text{Class} = -) * P(B = 1 | \text{Class} = -) * P(C = 0 | \text{Class} = -)) * P(\text{Class} = -) = \\ \frac{2+2}{5+4} * \frac{2+2}{5+4} * \frac{0+2}{5+4} * \frac{5}{10} = 0.022$$

Normalize:

$$P(\text{Class} = + | A = 0, B = 1, C = 0) = \frac{0.025}{0.025 + 0.022} = 0.53$$

$$P(\text{Class} = - | A = 0, B = 1, C = 0) = \frac{0.022}{0.025 + 0.022} = 0.47$$



5.

a. CPT

Polution = L	0.55
Polution = H	0.45

Smoker = True	0.35
Smoker = False	0.65

	Cancer = True	Cancer = False
Xray = pos	0.667	0.286
Xray = neg	0.333	0.714

	Cancer = True	Cancer = False
Dyspnoea = True	0.667	0.286
Dyspnoea = False	0.333	0.714

	Polution = L Smoker = True	Polution = L Smoker = False	Polution = H Smoker = True	Polution = H Smoker = False
Cancer = True	0.333	0.125	0.5	0.4
Cancer = False	0.667	0.875	0.5	0.6

b. $E = \{P=L, S=True, C=True, X=pos\}$

$$P(D = True | E) = \frac{P(D = True, E)}{P(E)} = P(P = L, S = True, C = True, X = pos, D = True) \\ = P(P = L) * P(S = True) * P(C = True | P = L, S = True) * P(X = pos | C = True) * P(D = True, C = True)$$

$$= 0.55 * 0.35 * 0.333 * 0.667 * 0.667 = 0.0285$$

$$\begin{aligned} P(D = False|E) &= \frac{P(D = False, E)}{P(X)} = P(P = L, S = True, C = True, X = pos, D = False) \\ &= P(P = L) * P(S = True) * P(C = True|P = L, S = True) * P(X = pos|C = True) * P(D = False, C = True) \\ &= 0.55 * 0.35 * 0.333 * 0.667 * 0.333 = 0.0142 \end{aligned}$$

$$P(D = True|E) = \frac{0.0285}{0.0285 + 0.0142} = 0.667 \rightarrow \text{Likely have dyspnoea}$$

$$P(D = False|E) = \frac{0.0142}{0.0285 + 0.0142} = 0.333$$

6. <https://github.com/fidelisprasetyo/CS5990/tree/hw4>